RISK IN SUPPLY CHAIN MANAGEMENT

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Abstract: The supply chain consists of many activities and organizations through which the suppliers' material moves up to the end buyer. During its movement, the materials pass a long way, starting from the raw materials suppliers, through the producers, operations, logistic centers, warehouses, transport companies, wholesalers, retailers, to the final buyers. Sometimes, the supply chains go beyond the end buyers, in the sense of recycling waste materials and their use as raw materials. Complexity and disintegration are emerging as major challenges in supply-chain risk management. It has become more difficult to identify risks as supply-chain operations have fallen into the hands of outside service providers, and are therefore less visible. The risks, their identification and impact depend on the position of the companies in the chain, and on the level of analysis they can carry out.

The paper will provide an analysis of the potential risks and models for risk management in the supply chain. The paper also highlights the key problems faced by supply chains during the pandemic, and some recommendations for overcoming them will be given.

INTRODUCTION

Supply chain management is the management of the flow of goods and services and includes all processes that transform raw materials into final products. It involves the active streamlining of a business's supply-side activities to maximize customer value and gain a competitive advantage in the marketplace (Jason, 2022).

Christopher (2011) defined SCM as a management of upstream and downstream relationships with suppliers and customers in order to deliver superior customer value at less cost to the supply chain as a whole.

Without mutual collaboration and coordination, the materials flow and obtaining the right product in the right time, right amount, right place to the right buyer, are in question.

The Supply-Chain Council defines Supply chain Management (SCM) as the effort in producing delivering a final product from the supplier's supplier to the customer's customer

A supply chain is the linkage of series of organizations with facilities, functions, processes, and logistics activities that are involved in producing and delivering a product or service. In the past, when firms manufactured in-house, sourced locally and sold direct to the customer, 'risk' was less diffused and easier to manage. With the advent of increased product/service complexity, and outsourcing of supply networks across international borders, risk is increasing and the location of risk has shifted through complex changing supply networks. Supply chains are becoming more complex, dynamically changing webs of relationships. Whilst this complexity arises from many sources, some key drivers are increasing product/service complexity, e-business, outsourcing, and globalization. Having in mind the importance of the chains and on the other hand the complexity and risks during the realization of the activities, the aim of the papers is to provide definitions for risks and risk classification in order for better identification.

The rest of the paper is organized as follows. Section 2 deals with some previous work on this topic. Section 3 gives theoretical overview and introduce briefly the readers with supply chain and supply chain management,. Section 4 provides some classification for risk that may occur in the supply chain. Section 5 gives a referent model for risk managing in supply chain. Section 6 gives an overview for supply chain during pandemic with COVID 19 and the last section concludes conclusion and recommends.

LITERATURE REVIEW

The effective supply chain management has become an important enabler to improve organization performance and valuable way of securing competitive advantage (Chirderhouse et al, 2003; Li et al, 2006). Supply Chain Risk Management (SCRM) is an area that has recently been receiving a great deal of interest from academics and practitioners. SCRM is believed to be in an emerging and promising new field by researchers (Sodhi et al., 2012). The supply chain risks are a set of obstacles to initiatives taken in the context of moving products from their place of production to the final consumer. Risks can be defined as a possible variation in the distribution of supply chain results, their likelihood and subjective values or a break in flow between the components of the supply chain (Lavastre & Spalanzani, 2010). Earlier literature consider risks in relation to supply lead time reliability, price uncertainty, and demand volatility which lead to the need for safety stock, inventory pooling strategy, order split to suppliers, and various contract and hedging strategies (Tang, 2006). According to Chopra and Sodhi (2004), the supply chain risks could be in the form of delays of materials from suppliers, large forecast errors, system breakdowns, capacity issues, inventory problems, and disruptions.

THEORETICAL OVERVIEW FOR SUPPLY CHAIN AND SUPPLY CHAIN MANAGEMENT

The supply chain consists of many activities and organizations through which the suppliers' material moves up to the end buyer.

The supply chain encompasses functions that are linked to satisfy the customers' needs. Accordingly, these functions include: new product development, marketing, manufacturing operations, distribution, finance and customer services. (Regodić, 2014)

At the same time, you should have in mind that each product has a unique supply chain, that some chains are short and simple, while others are long and complex.

The cooperation in the management and the trust among the stakeholders are in the supply chains focus. Without mutual collaboration and coordination, the materials flow and obtaining the right product in the right time, right amount, right place to the right buyer, are in question
So, the supply chains function on the following principles:

- Understanding the level of the end consumers' demands;

- Making decision where to place the stocks along the chain and how long to be kept at each place;
- Development of adequate policies and procedures for the supply chains management as the only entity.

The supply chain management is a complex process that consists of several components. The main components of the supply chain management, as it can be seen from the figure itself, are:

- The chain structure – the supply chain members to which the processes will be connected;
- Business processes – which processes should be connected with all chain participants. In fact, those are the activities that produce specific results which present a value for the consumers;
- Management components – which integration and management level should be applied for connecting the supply chain elements.

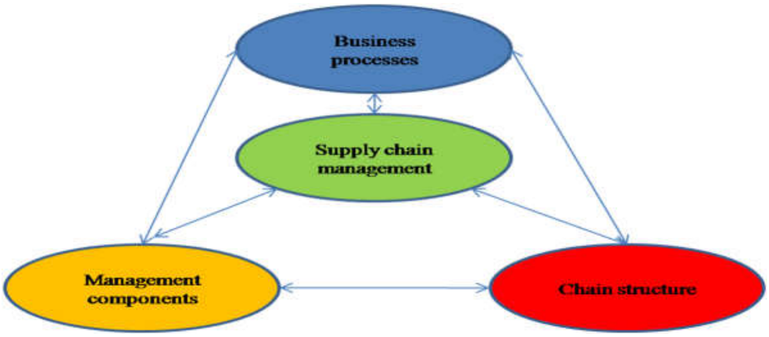

Figure 1. Supply chain components

RISK IN SUPPLY CHAIN RISK

The risk in the supply chain is a measure for the exposure to a risky case and generally two categories of cases are implied. In the first category are cases connected with natural or other disasters which are not a result of organized implementation of some business activities. In the second category are various cases realized by individual members in and out of the chain, directed to the business aims. What is characteristic for the mentioned activities, is the fact that during their planning, organizing, realizing and controlling, the risks that can occur, are not taken into the account, and because of that, such a situation is a fertile soil for the risks occurrence and growth in the supply chain.

The concept of risks in the supply chain can be defined as an exposure to risky cases that have negative influence on the chain operability and on its performances, such as the level of the service for the user, the responsiveness, costs, etc.

The risky cases that can occur in the supply chain are different, and usually they are a consequence of two kinds of interaction, i.e.: the first one that occurs between the members in the supply chain and its environment, and the second one, an interaction between the individual members of the supply chain.

The management of the risks in the supply chain is multi-dimensional phenomenon with a wide spectrum of circumstances that influence it and very complex procedures and way of implementation.
According to Norrman and Lindroth (2004), the management of the risks in the supply chain is a process of managing the risks in the sense of treating the uncertainty related to the logistic activities.

While Waters (2007) mentions several principles on which the management of the risks in the supply chain is based, i.e.:

- The first principle is related to the balance which should be realized between the desire for higher effectiveness and the growing vulnerability of the supply chain. Usually, when the methods for increasing the effectiveness are applied, the risks on which the applied method will impact are not taken into the account.
- The second principle is proactivity. Waiting for realization of a risky case is not good for the company, and because of that the management of the risks has to ensure the company to be ready for occurrence of the risky case; therefore, the risk management should be based on identification, analysis and response.
- The third principle indicates that the risks management in the supply chain is cyclic, not linear process; in other words, that means the process never ends, but its advancement is constantly planned, having in mind the new changed conditions. One way for advancing the management of the risks is the well-known technique PDCA (plan-do-sheck-act).

TYPES OF RISKS

Usually there are two key types of supply chain risk management, both internal and external.

External risks come from outside and are not controlled by the supply chain, and they are often than the internal.

- demand risks - caused by unpredictable or misunderstood customers or end-user claims.
- supply risks - caused by any interruptions in the flow of products, raw materials or parts, within the supply chain.
- risks from the environment - usually related to economic, social, governmental and climatic factors, including the threat of terrorism.
- business risks - caused by factors such as the financial or managerial stability of the supplier or the buying and selling of suppliers.

While internal risks create better opportunities because they can be affected by the supply chain, these include:

- production risk - caused by disruption of internal operations or processes
- business risk - caused by changes in key personnel, management, reporting structures or business processes, such as the way customers communicate with suppliers and customers
- planning and control risks - caused by inadequate assessment and planning, which is ineffective management
- mitigation risks and unforeseen situations - caused by unobtrusiveness (or alternative solutions) in case something goes wrong
- cultural risks - caused by the business's cultural tendency to hide or delay negative information. Such businesses generally react more slowly when they are hit by unexpected events

The more detailed information about the risk than arise as result of new trends in supply chain management is given in the table 1.

Table 1. Characteristic risks as a consequence of new trends in supply chain management

Trend	Consequence of supply chain flows	Risks due to changes in flows
Agility	Increasing the flows dynamics	Increase errors
		Increasing stocks that are old and moving slowly (non-current stocks)
Product customization	Increased complexity of flows	Increasing complexity increases the risks of flows interruptions
Globalization	Increased geographical coverage of the market, procurement and sales	Increased distance with multiple transport links affects increased exposure to potential outage
		Increasing cultural differences increase the risk of misunderstanding
		Complex operation is due to complex communication due to multiple time zones
		Longer distances increase the cost of direct personal contact
Delivery time	Short delivery time	Less time to react when errors occur
Lean	Low stock reliability	Less opportunity to react to the interruption of the use of security supplies
	Short delivery time	Less opportunity to react to the interruption of the use of the time buffer
Outsourcing	More connections in the supply chain	Increased complexity and potentially longer supply chain increases the risk of disruption
Product life	Reducing the life of the product	Decreased chain stability increases the risk of disruption to logistics
Reducing the suppliers number	Single or dual sourcing	The reduced number of alternatives increases the negative consequences in case something goes wrong with the selected supplier

Source: Maslarić 2014

The mentioned trends and the consequences of their application can cause numerous risks due to which there will be an abruption in the logistics flows, which will affect the supply chain. On the other hand, with the traditional methods for the risks treating, such as safe stocks and reserves at the time of delivery, the appearance of the mentioned trends is decreased.

MODELS FOR RISK MANAGEMENT IN THE SUPPLY CHAIN

The models of processes for the risk management in the supply chain present a systematic application of policies, procedures and the risk monitoring in the supply chain.

The process models applied for risk management in supply chains are very similar to the models applied in the context of quality management.

The complexity of the proposed or applied models of risk management processes in supply chains, in terms of the number of elements that make up a given model is different depending on the author who proposed the model.

Table 2 shows some of the models of risk management processes in supply chains according to the elements that make them up.

Table 2: Models of risk management

elements/ authors	Context defining	Risk identification	Risk analysis	Risk assessment	Risk aapraisal	Risk evaluation	Risk treatment	Decision-making for measures and implementation	Monitoring and control
Paulsson,2007	*	*	*		*	*	*		*
Waters,2007	*	*	*				*		*
Norrman & Jansson, 2004		*			*		*		*
Tuncel & Alpan,2010		*		*	*			*	*
Blome & Schoenherr,2011		*	*				*		*

Source: Paulsson, Waters, Norrman &Jansson, Tuncel & Alpan, Blome & Schoenherr

The data in the table show that different authors state different models for risk management in the supply chain, but it's what is important to note that risk must be identified first and then measures must be taken to treat it.

SUPPLY CHAIN AND RISKS DURING COVID 19 PANDEMIC

The COVID-19 pandemic has created a shock for supply chains around the world. Some products have seen a sudden rise in demand, while it has dropped rapidly for others. Lockdowns, including closing businesses and remote working, in countries around the world, have hindered the flow of raw materials, people, and finished products in the supply chain.

Here are some tips for managing supply chain during pandemic: (François, 2022)

- gather your leadership team. This team should meet daily to revise your sales and operations plans and be ready to adjust if any issues arise. Put your people and employee at the centre of this.
- determine the level of demand for every product. Key customers need to be contacted to determine their needs in the coming weeks.
- Get information about current inventory levels for various types of goods
- Analyze inbound and outbound supply chain risks.Inbound analyze include: sourcing key suppliers, sourcing contract renegotiations, transport, and inbound shipping, storage / receiving. Outbound analyze include: warehousing, packaging, distribution mode, transport and outbound shipping
- Make plans and strategy. Having identified your risks, you should look for ways to mitigate them. Prioritize issues that have a higher risk score.
- Communicate, and think long term.

CONCLUSION

Supply chains are important for economic activities realization, but they are not often highly visible. These networks of buyers and suppliers are ultimately tasked with delivering raw materials, intermediate goods, and, eventually, end products to consumers and businesses around the world. For the most part, supply chains operate efficiently in the background, out of the sight of end consumers. During the realization of the activities from the supplier to the final buyer, numerous risks may appear which were subject to identification in this paper.

To overcome the consequences of risks in the supply chain each participant should anticipate and mitigating future risk before it is realised in an effort to improve supply chain resilience and to identifying critical supply chains and localising them to ensure supply.

It is also necessary to note that COVID-19 pandemic has changed the business environment for many organizations around the globe, and has highlighted the importance of being able to react, adapt and set up crisis management mechanisms in order to weather situations of uncertainty. As the acute restrictions and lockdowns created many urgent situations that required immediate attention in the early days of the pandemic, many companies have now begun to move to a "recovery mode" and have started planning for the longer term. As companies seek to strengthen operations and business resilience, the importance of supply chain resilience and risk management is more apparent than ever.

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